

STORLIEN-STORVALLEN, Sweden

WMO: 022070

Lat: 63.283N

Lon: 12.122E

Elev: 1913

StdP: 13.71

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -12.3	(c) -5.8	(d) -16.5	(e) 2.4	(f) -12.1	(g) -9.8	(h) 3.5	(i) -4.9	(j) 40.9	(k) 23.6	(l) 34.6	(m) 28.3	(n) 2.0	(o) 120	(p) 0.764

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.7	(c) 73.4	(d) 57.9	(e) 69.7	(f) 56.8	(g) 66.1	(h) 54.9	(i) 60.6	(j) 69.5	(k) 58.6	(l) 66.8	(m) 56.5	(n) 63.8	(o) 6.9	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.9	74.1	63.9	54.8	68.6	61.5	52.8	63.7	59.9	27.7	69.4	26.3	66.7	24.9	63.9	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.6	23.4	19.9	-18.2	78.1	7.2	4.0	-23.4	80.9	-27.6	83.2	-31.6	85.4	-36.8	88.3
			-19.0	63.9	6.9	3.3	-24.0	66.2	-28.1	68.1	-32.0	69.9	-37.0	72.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.4	20.9	20.9	24.8	31.8	40.2	48.0	53.7	52.1	45.6	35.9	26.9	22.7
	DBStd	14.46	10.51	10.86	8.62	6.65	7.24	6.84	6.90	6.13	5.97	7.14	9.48	11.14
	HDD50	5702	902	815	783	547	318	119	39	46	156	438	692	847
	HDD65	10820	1367	1235	1248	997	769	511	355	401	582	901	1142	1312
	CDD50	364	0	0	0	0	14	58	154	112	24	2	0	0
	CDD65	7	0	0	0	0	0	1	5	1	0	0	0	0
	CDH74	93	0	0	0	0	3	7	77	6	0	0	0	0
	CDH80	11	0	0	0	0	0	0	11	0	0	0	0	0
Wind	WSAvg	8.0	9.7	9.6	10.0	7.7	7.1	7.0	5.9	6.2	7.4	8.3	8.1	9.0
Precipitation	PrecAvg	34.8	3.0	2.5	2.4	1.7	2.2	3.3	4.1	3.8	3.5	3.0	2.6	2.7
	PrecMax	44.9	6.3	6.0	5.0	3.9	3.7	5.0	7.3	6.5	7.8	4.6	6.5	5.7
	PrecMin	23.6	0.4	0.2	0.8	0.6	1.1	1.1	1.4	1.4	1.0	1.3	0.2	0.6
	PrecStd	4.3	1.6	1.2	1.3	0.9	0.6	1.0	1.5	1.4	1.9	0.9	1.8	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	40.6	42.5	53.4	72.1	74.5	81.5	74.4	66.9	57.6	47.6	41.7
		MCWB	35.3	36.2	36.8	43.2	53.9	56.5	61.6	58.5	53.9	49.1	43.6	39.0
	2%	DB	36.9	37.4	39.5	48.2	64.3	68.8	75.4	70.4	61.9	51.9	43.7	39.2
		MCWB	34.3	34.8	35.9	40.2	50.9	54.8	60.1	57.0	52.7	46.2	40.6	37.1
	5%	DB	35.2	35.2	37.6	44.2	57.7	64.6	71.6	66.9	58.6	48.9	40.5	37.3
		MCWB	33.3	33.0	34.7	38.0	47.9	52.6	58.6	55.8	51.0	44.3	37.6	35.1
	10%	DB	33.2	33.4	35.6	41.4	52.4	60.8	67.4	63.5	55.4	46.2	38.0	35.1
		MCWB	31.3	31.5	33.0	36.7	44.7	51.1	56.3	54.9	49.6	42.3	35.3	33.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.8	37.5	38.5	44.2	55.6	60.3	64.7	61.2	56.5	50.7	44.5	39.7
		MCDB	37.9	39.4	39.8	51.1	68.6	68.8	77.0	68.9	62.4	54.9	47.3	40.8
	2%	WB	34.9	35.3	36.6	40.9	51.8	56.4	62.1	59.3	53.9	46.9	41.1	37.4
		MCDB	36.3	37.0	38.6	46.2	62.5	63.8	72.4	67.2	59.7	50.9	43.2	38.8
	5%	WB	33.4	33.6	34.9	38.8	48.5	53.9	59.8	57.2	52.0	44.7	38.0	35.3
		MCDB	35.1	34.9	37.2	43.3	57.3	62.4	68.8	64.2	57.6	48.4	40.1	37.0
	10%	WB	31.6	31.6	33.3	37.1	45.3	51.8	57.8	55.5	49.9	42.6	35.6	33.3
		MCDB	33.0	33.1	35.4	40.9	51.7	59.8	66.1	61.7	54.6	45.9	37.7	34.8
Mean Daily Temperature Range	5% DB	MDBR	10.2	10.2	11.0	12.2	13.4	14.4	16.7	15.5	12.9	9.6	9.4	10.0
		MCDBR	8.7	8.6	10.5	15.9	24.9	23.1	25.9	21.8	18.8	12.9	10.0	9.1
		MCWBR	7.7	7.6	8.0	10.3	13.7	11.9	13.1	11.3	11.4	8.6	8.4	8.5
	5% WB	MCDBR	8.2	8.4	9.7	13.8	23.7	20.0	23.2	17.8	17.3	11.9	9.7	8.7
		MCWBR	7.6	8.0	7.9	9.1	13.7	10.9	12.2	10.2	11.0	8.7	8.5	8.4
Clear-Sky Solar Irradiance	taub		0.197	0.259	0.288	0.316	0.327	0.326	0.335	0.324	0.300	0.280	0.207	0.182
	taud		2.012	2.216	2.232	2.268	2.411	2.492	2.481	2.505	2.558	2.449	2.073	1.954
	Ebn at Noon		141	219	257	274	282	285	278	272	258	214	133	73
	Edh at Noon		10	19	28	33	32	30	30	26	20	15	9	4
All-Sky Solar Radiation	RadAvg		55	228	654	1234	1468	1542	1482	1130	679	295	82	27
	RadStd		7	33	76	97	102	144	163	116	66	36	11	2

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-256	-320	N/A	N/A

Nomenclature: See separate page